Project Title:

Smart Retail Sales Analysis & Demand Forecasting

Abstract

Retail businesses generate large volumes of sales data every day, making it essential to analyze this information for effective decision-making and business growth. The **Smart Retail Sales Analysis & Demand Forecasting system** is developed to analyze historical retail sales data, identify sales patterns, and forecast future product demand using data analytics and machine learning techniques. The system processes a retail sales dataset containing information such as product details, sales transactions, customer information, order dates, quantities sold, discounts, profits, and regional sales. Data preprocessing techniques including data cleaning, handling missing values, duplicate removal, feature engineering, and data transformation are applied to ensure data quality and reliability. Exploratory Data Analysis (EDA) is performed to discover sales trends, customer purchasing behavior, product performance, seasonal demand patterns, and regional sales distribution through visualizations such as bar charts, line graphs, scatter plots, histograms, and heatmaps. An interactive dashboard is developed using Power BI to present key performance indicators and business insights in an intuitive manner. Furthermore, machine learning models are trained on the processed dataset to forecast future product demand, enabling better inventory planning and sales forecasting. A user-friendly web application is developed to allow users to input retail data and obtain demand predictions from the trained model. The proposed system demonstrates how data analytics, visualization, and predictive modeling can support smarter business decisions, improve inventory management, and enhance overall retail performance.

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